



CS 431/631 Introduction to Big Data

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Instructor and TA

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please make an appointment before you come*

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MEET WITH EACH OTHER

Let's meet with each other!

1. Your **name**? Brian
2. Which **program**, which **year**? CSE, 4th
3. Professional **interests**? Cyber defense
4. Others (personality, hobbies, stories...) Guitar, gym

BIG DATA MINOR

I. PROGRAM REQUIREMENTS (18 UNITS)

The minor requires a minimum of 18 units from the courses listed below. No more than nine of the units can also count towards a major or other degree requirements.

A. REQUIRED COURSES (6 UNITS)

- [CS 431 - Introduction to Big Data](#) (3 units) ✓
- [CS 457 - Database Management Systems](#) (3 units) ✓

B. GROUP I (6-12 UNITS)

Choose two to four courses from the list of Group I courses.

- [CS 422 - Introduction to Machine Learning](#) (3 units)
- [CS 433 - Data Intensive Computing](#) (3 units)
- [CS 436 - Big Data Systems](#) (3 units)
- [CS 458 - Introduction to Data Mining](#) (3 units)
- [CS 461 - Statistical Methods in Bioinformatics](#) (3 units)
- [CS 487 - Fundamentals of Deep Learning](#) (3 units)
- [STAT 452 - Introduction to Regression and Linear Models](#) (3 units)
- [STAT 453 - Categorical Data Analysis](#) (3 units)

More details: https://catalog.unr.edu/preview_program.php?catoid=47&poid=130555

BIG DATA MINOR

C. GROUP II (0-6 UNITS)

Any remaining courses can be selected from the list of Group II courses.

- [CS 415 - Parallel Computing](#) (3 units)
- [CS 442 - Cloud Computing](#) (3 units)
- [CS 477 - Analysis of Algorithms](#) (3 units) ✓
- [CS 479 - Pattern Recognition](#) (3 units)
- [CS 482 - Artificial Intelligence](#) (3 units)
- [IS 482 - Applied Data Science](#) (3 units)
- [MATH 420 - Mathematical Modeling](#) (3 units) CO13, CO14
- [STAT 445 - Introduction to Statistical Computing](#) (3 units)
- [STAT 467 - Statistical Theory](#) (3 units)

COURSE SCHEDULE

- ***Big Data Overview***
- ***Fundamentals in Big Data Analytics***
Data Collection and Exploration, Basic Data Processing and Analysis
- ***Intro to Big Data Machine Learning***
Machine Learning Overview, Learning Types, Build and Train Machine Learning Models
- ***Essential Big Data Systems and Tools***
MapReduce, Spark, Machine Learning Systems
- ***Optional Selected Big Data Topics***
- ***Mid-term***
- ***Final Project Presentation***

COURSE OVERVIEW

1. total 16 weeks, (31-2=29) classes.
2. 22 lectures by instructor, 6 for project presentation, 1 for mid-term.
3. 0 required textbooks, 0 final exam, 0 homework. yay
4. Lightweight workload: quizzes + mid-term + project
↓ ↓ ↓
daily based on seem long
 quizzes?

POINT DISTRIBUTION

- Quiz 30%
- Midterm Exam 20%
- Project
 - Proposal 10%
 - Final Report 25%
 - Presentation 15%

GRADING CRITERIA, SCALE, AND STANDARDS:

A: 93% - 100%
A-: 90% - 92.9%
B+: 87% - 89.9%
B: 84% - 86.9%
B-: 80% - 83.9%
C+: 77% - 79.9%
C: 74% - 76.9%
C-: 70% - 73.9%
D+: 67% - 69.9%
D: 64% - 66.9%
D-: 60% - 63.9%
F: <60%

■ Course Policies:

- Students are required to follow COVID policies.
- Students are expected to attend, and be on time, for every class. This demonstrates professionalism and consideration for your fellow students and your Instructor. Students who miss class and/or are late for class may experience an impact on their grade by missing classroom activities and/or quizzes.
- Students are expected to turn in all assigned materials in a timely manner.
- Students are expected to demonstrate professionalism and courtesy by either silencing or turning off all cell phones and/or other alarm or audible indicator devices.
- The Instructors reserve the right to add to, and/or modify any of the above policies as needed to maintain an appropriate and effective educational atmosphere in the classroom and the laboratory. In the case that this occurs, all students will be notified in advance of implementation of the new and/or modified policy.

MISSING CLASS

*Email me **before** class unless it is not feasible
(justify the situation)*

*For COVID-19 or health related, medical proof is
NOT required*

MISSING CLASS

*Make up for missing classes
(You can submit at most 3 reviews.)*

Write 1 paper/book/article evaluation for each missing class (at least one page for each)

- 1. Summary (problem, method, novelty, evaluation)*
- 2. Pros (at least 3)*
- 3. Cons (at least 3)*
- 4. Comments (what can be improved, different thoughts)*

QUIZZES

In class quizzes

- 1. How: WebCampus by using access code*  of + "bigdata"
- 2. When: at the end of class*
- 3. What: content in this class and previous classes*

MID-TERM

Mid-term

- 1. How: in-class on paper*
- 2. When: 3/30 (tentative)*
- 3. What: quizzes + things we talked in class (True or False, Multiple Choices)*

COURSE PROJECT

Project

- A selection of topics will be provided *R11*
- You can come up with your own topics *X*
- Project team: optional, no more than 3 members
- Project proposal
- Project presentation
- Final Project report

*matt, me,
adrian*

More details will be available on WebCampus later

COURSE PROJECT

Project Proposal Structure

- 1. Problem*
- 2. Tasks to finish*
- 3. Tech/Tools/Systems (or paper list for survey project)*
- 4. Implementation/Evaluation Plan (justification)*
- 5. Timeline and milestones*

COURSE PROJECT

Final Project Report Structure (hands-on)

- 1. Abstract*
- 2. Introduction*
- 3. Background/motivation*
- 4. Approach*
- 5. Evaluation*
- 6. Discussion (optional)*
- 7. Related Work*
- 8. Conclusion and future work*
- 9. References*

COURSE PROJECT

Final Project Report Structure (survey)

1. Abstract
 2. Introduction
 3. Background/motivation
 4. Sum and compare literatures (≥ 10 papers)
 5. Evaluation (optional)
 6. Discussion (optional)
 7. Conclusion and future work
 8. References
- not relevant to me :)*

COURSE PROJECT

Presentation Structure

- 1. Background (big / small background - timing)*
- 2. Motivation (why)*
- 3. Problem Definition (what)*
- 4. Methodology (solution/answer)*
- 5. Evaluation (experiment setup, results, observations)*
- 6. Conclusion (sum, **take away message**)*

COURSE PROJECT

Good Presentation

1. *Problem Definition and Take Away!!!*
2. *NO full/long sentence, font ≥ 18*
3. *Images and animations*
4. *Less is more (how much audience get)*
5. *Practice! Practice! Practice!*
6. *Interaction & time*

UNDER VS GRAD

nile

Difference: Under VS Grad

- 1. Grading: less expectation of everything for Under.*
- 2. Project: survey is an option for Under.*

THANK YOU!

Questions?